THz-TDS Time-Domain Pulse: Physics Parameters → Waveform Features
Single-cycle pulse with material-dependent delay and absorption
Polymer (n=1.55, α=8.0 cm^{-1})
Ceramic (n=1.75, α=18.0 cm^{-1})
Composite (n=1.65, α=12.0 cm^{-1})
Pre-pulse Region
Baseline noise &
experimental drift
Absorption Tail
$A(t) \propto \exp(-t/\tau)$
$\tau \propto 1/\alpha$
Ceramic: fastest decay
Electric Field Amplitude (a.u.)
1.5
1.0
0.5
0.0
−0.5
Time (ps)
−4
−2
0
2
4
6
8
Δt = 1.00 ps